NIRMAL KHEDKAR

Software Engineer | Reactive Systems & Cloud-Native Architecture | Ex-Visa | MS CS @ CU Boulder
linkedin.com/in/nirmalhk7 ◇ nirmalhk7.com ◇ github.com/nirmalhk7
nirmalhk7@gmail.com

RELEVANT INDUSTRY EXPERIENCE

ZOLL Data Systems May 2025 - Aug 2025
Software Engineer Intern

- - Built a C#/.NET scheduling engine for EMT dispatch with a modular data layer that other components can reuse, improving route assignment efficiency 30% via priority-based route-sorting logic.
- Integrated the AI optimization engine with production telemetry, achieving sub-second EMT assignment latency in emergency dispatch.

Visa Inc. Jun 2022 - Jul 2024
Fullstack Software Engineer

- Migrated latency-critical Java/Spring Boot services from thread-per-request (Spring MVC) to non-blocking event-loop concurrency (**Project Reactor**), increasing per-instance throughput by 30% under sustained load.
- Investigated cross-functional production incidents using Splunk and Grafana to diagnose latency and throughput bottlenecks on APIs operating to a **99.9999% uptime SLA**.
- Standardized team's DevSecOps protocols by integrating **SAST/DAST** and **SonarQube** into Jenkins CI pipelines, reducing deployment-blocking vulnerabilities by 75%.
- Refactored legacy Spring Boot microservices to improve modularity and performance across core payment APIs.

Twizzr.com Oct 2019 - May 2021
Founding Software Engineer (Part Time / Undergrad)

- Designed and shipped Node.js backend services as **founding engineer**; refactored the legacy codebase into modular packages and established the team's CI/release conventions.
- Set up Jenkins CI/CD with ESLint and semantic-release, eliminating manual release steps and enforcing style on every PR.

Visa Inc. May 2021 - Jul 2021
Software Engineering Intern

- Analyzed Visa Developer Platform usage data to surface trends across routes, APIs, products, and clients.
- Consolidated fragmented data streams into a unified RESTful backend and React dashboard, streamlining internal querying and data visualization for product stakeholders.
- Fine-tuned system throughput by building a dynamic Throttling and Load Prediction module, achieving a 15% gain in overall performance.

EDUCATION

University of Colorado - Boulder Aug 2024 - May 2026
Master of Science - Computer Science

National Institute of Technology Karnataka Jul 2018 - May 2022
Bachelor of Technology - Information Technology

TECHNICAL SKILLS

Backend & Distributed Systems: Java (Spring Boot, Project Reactor, R2DBC), Node.js (NestJS, Express), Python, C#/.NET, GraphQL, gRPC, RabbitMQ, Redis, Kafka.
Cloud & Infrastructure (IaC): Google Cloud (GCP), Kubernetes, Terraform (IaC), Docker, Jenkins CI/CD, Linux/Unix.
Data & Observability: NoSQL, MySQL, PostgreSQL, MongoDB, Spark, InfluxDB, Splunk, Prometheus, Grafana.
Frontend & Mobile: TypeScript, JavaScript, ReactJS, Redux, Cypress, NextJS, AngularJS, React Native, Gatsby, Android (Kotlin), HTML5/CSS3.
Engineering Practices: JVM Tuning, Microservices Architecture, High-Availability Systems, SAST/DAST Security, DevSecOps, Gen AI, Agile/Scrum.

PROJECTS

Preemptive Kubernetes Autoscaler Kubernetes, InfluxDB, Time-series prediction

- Developed a predictive Kubernetes autoscaler leveraging Kubernetes REST APIs, Prometheus metrics, and InfluxDB historical telemetry to preemptively scale deployments horizontally/vertically; packaged as a Helm chart.

Globally Distributed Consensus Based Chat Service Raft, IRC, Distributed Systems

- Built a chat backend using Raft consensus with regional partitioning and Chord-based DHT lookup for routing channels to their owning region; implemented leader election, log replication, and total message ordering within partitions.

Self-Healing Automated Home Infrastructure GitOps, K8s, LLMs

- Built a GitOps-driven home infrastructure on Terraform and Ansible with immutable deploys, plus an Agentic system (Openclaw) that parses system failures and triggers recovery processes automatically.

Custom Shell with Process Management C, Unix System APIs, POSIX

- Developed a custom Unix shell in C supporting command execution, I/O redirection, and piping using fork/exec syscalls, featuring job control and signal handling for robust process management.

Driver Attentiveness Detection Image Analysis, Computer Vision

- Engineered a PyTorch/OpenCV computer vision pipeline analyzing dash-cam footage for driver drowsiness using CNN-based feature extraction; achieved 94% accuracy in detecting predefined inattentiveness indicators in test datasets.

ACADEMIC ACHIEVEMENTS

Vehicle Re-identification Using CNN Jan 2023

Research Paper - ICADCML 2023

- The paper describes the usage of CNNs with vision-based techniques to identify vehicles across multiple cameras.

NIT Conclave 2019 Hackathon

2nd Position for College

- Participated in the hackathon team developing a full-stack social network platform for researchers.

RELEVANT CERTIFICATIONS & SPECIALIZED TRAINING

GCP Fundamentals: Core Infrastructure Google Cloud

CKAD (Kubernetes Application Developer) Crash Course O'Reilly

AI Pair Programming with GitHub Copilot LinkedIn Learning

Prompt Engineering: How to Talk to the AIs LinkedIn Learning

Getting Started with Go (Golang) UC Irvine